Identity-Based Encryption and CCA Security

CS 171: Cryptography

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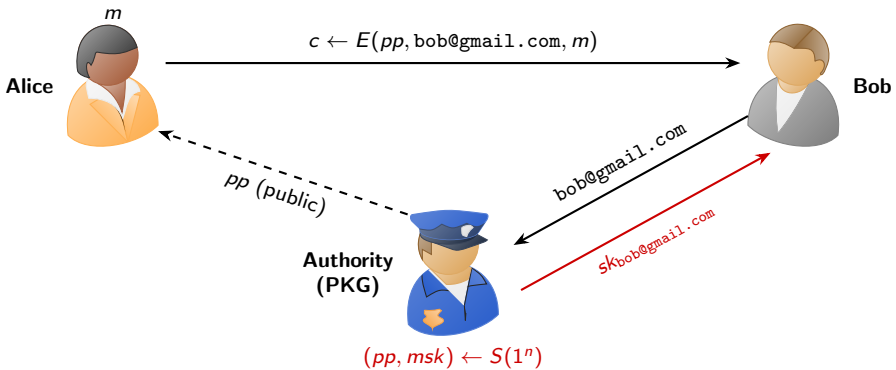
Overview

1. Identity-Based Encryption (IBE)
2. Bilinear Groups and the DBDH Assumption
3. Digital Signatures from IBE
4. CCA Security from IBE
5. Summary

Identity-Based Encryption (IBE) [Shamir84]

Idea

The **identity** of the recipient is the public key (e.g. an email address). No need to look up a separate public key for each recipient.



IBE: Syntax [Shamir84]

Definition 1 (Identity-based encryption)

An IBE scheme consists of four algorithms (S, K, E, D):

- $S(1^n) \to (pp, msk)$: **public parameters** pp and **master secret key** msk.
- $K(msk, id) \to sk_{id}$: the secret key for identity id.
- $E(pp, id, m) \to c$: encrypt m using pp and the identity id.
- $D(sk_{id}, c) \to m$: decrypt c using sk_{id}.

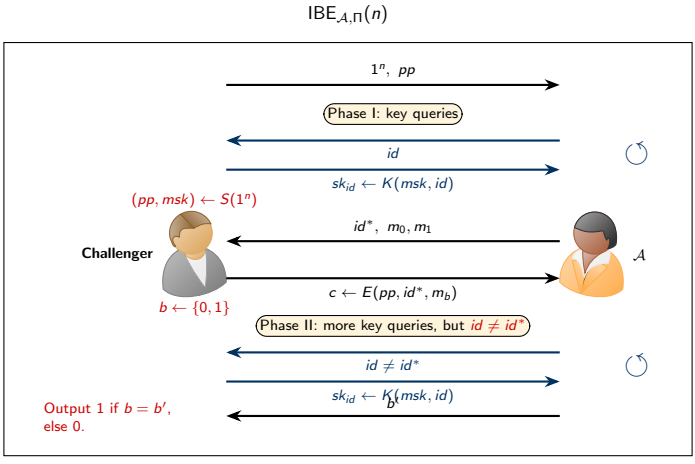
Correctness

For all id and m: if $(pp, msk) \leftarrow S(1^n)$ and $sk_{id} \leftarrow K(msk, id)$, then $D\big(sk_{id},\ E(pp, id, m)\big) = m$.

Security of IBE [BF01]

- The attacker has access to keys for **any number of identities of its choice**.
- The attacker still **cannot break security** for any *other* identity.

$\mathrm{IBE}_{\mathcal{A},\Pi}$: Challenger vs. Adversary



Security of IBE [BF01]

Game $\mathrm{IBE}_{\mathcal{A},\Pi}(n)$

1. $(pp, msk) \leftarrow S(1^n)$. $\mathcal{A}$ is given 1^n, pp, and oracle access to $K(msk, \cdot)$.
2. $\mathcal{A}$ outputs an identity id^* and m_0, m_1 with $|m_0| = |m_1|$.
3. $b \leftarrow \{0,1\}$, $c \leftarrow E(pp, id^*, m_b)$ is given to $\mathcal{A}$, which can make more key queries, and outputs b'.
4. Output 1 if $b' = b$ and $\mathcal{A}$ **never** queried $K(msk, id^*)$; else 0.

Definition 2 (IBE security)

Π is **secure** if for all PPT $\mathcal{A}$: $\quad \Pr\left[\mathrm{IBE}_{\mathcal{A},\Pi}(n) = 1\right] \leq \frac{1}{2} + \mathrm{negl}(n)$.

Selective Security

Game $\mathrm{IBE}^{\mathrm{sel}}_{\mathcal{A},\Pi}(n)$

The same as $\mathrm{IBE}_{\mathcal{A},\Pi}(n)$, except that $\mathcal{A}$ **commits to** id^* **at the very start**, after 1^n and **before** seeing pp:

1. $\mathcal{A}$ is given 1^n and outputs id^*.
2. $(pp, msk) \leftarrow S(1^n)$; $\mathcal{A}$ gets pp and key queries for $id \neq id^*$; ... (as before)

- A **weaker** notion: easier to achieve, and easier to prove.
- In a reduction, $\mathcal{B}$ knows id^* *before* it has to produce pp — and can plant its challenge there.

Bilinear Groups

- **High level**: groups where **CDH is hard** but **DDH is easy**.
- A group $\mathbb{G}$ of prime order q with generator g, with a **bilinear map** $e : \mathbb{G} \times \mathbb{G} \to \mathbb{G}_T$:
  - if g generates $\mathbb{G}$, then $e(g, g)$ generates $\mathbb{G}_T$;
  - $\forall\, a, b \in \mathbb{Z}_q$: $\quad e(g^a, g^b) = e(g, g)^{ab}$.
- **DDH is easy: how?** (g^a, g^b, h) is a DDH tuple iff $e(g^a, g^b) = e(g, h)$.
- **CDH is hard: how?** Cannot prove! Assumed, as no attacks are known.

Decisional Bilinear Diffie–Hellman (DBDH) Assumption

Game $\mathrm{DBDH}_{\mathcal{A},\mathcal{G}}(n)$

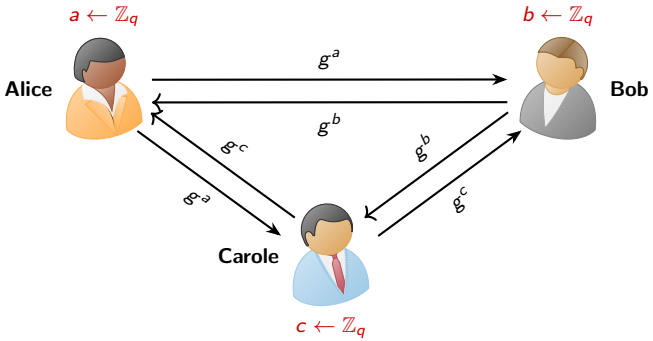
1. $(\mathbb{G}, \mathbb{G}_T, q, g, e) \leftarrow \mathcal{G}(1^n)$; $a, b, c, z \leftarrow \mathbb{Z}_q$, $\beta \leftarrow \{0,1\}$.
2. If $\beta = 0$ set $T := e(g,g)^{abc}$ (real); if $\beta = 1$ set $T := e(g,g)^z$ (uniform).
3. $\mathcal{A}$ is given 1^n, the group description, and $\big(g^a, g^b, g^c, T\big)$; it outputs β'.
4. Output 1 if $\beta' = \beta$, and 0 otherwise.

Definition 3 (DBDH)

DBDH is **hard relative to** $\mathcal{G}$ if for all PPT $\mathcal{A}$: $\Pr\big[\mathrm{DBDH}_{\mathcal{A},\mathcal{G}}(n) = 1\big] \leq \frac{1}{2} + \mathrm{negl}(n)$.

- Even given g^a, g^b, g^c, the value $e(g,g)^{abc}$ looks uniform in $\mathbb{G}_T$.

Three-Party Non-Interactive Key Exchange [Joux]



Shared key (one round!)

$k = e(g,g)^{abc}$: Alice computes it by raising $e(g^b, g^c)$ to the power a:

$$k = e(g,g)^{abc} = e(g^b, g^c)^a = e(g^a, g^c)^b = e(g^a, g^b)^c.$$

By DBDH, k is indistinguishable from uniform to an eavesdropper who sees g^a, g^b, g^c.

IBE Construction [BF01]

Construction

Let $H : \{0,1\}^* \to \mathbb{G}$ be a hash function.

- $S(1^n)$: $a \leftarrow \mathbb{Z}_q$; $pp = mpk := g^a$, $msk := a$.
- $K(msk, id)$: output $sk_{id} := H(id)^a$.
- $E(mpk, id, m)$, $m \in \mathbb{G}_T$: $r \leftarrow \mathbb{Z}_q$; output $c_0 := g^r$, $c_1 := m \cdot e\big(mpk, H(id)\big)^r$.
- $D(sk_{id}, (c_0, c_1))$: output $c_1 / e(c_0, sk_{id})$.

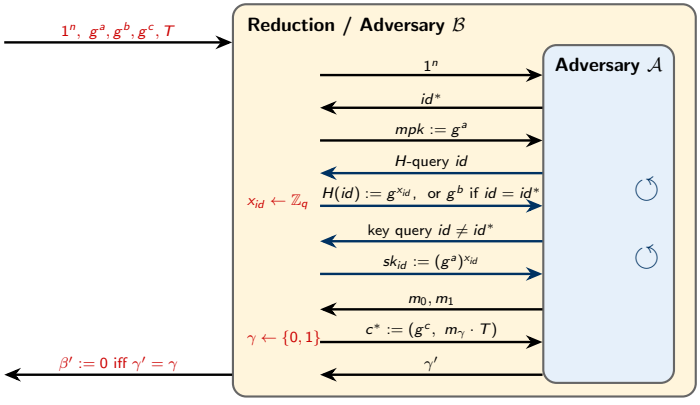
- **Correctness**: $e(c_0, sk_{id}) = e(g^r, H(id)^a) = e(g, H(id))^{ar} = e\big(mpk, H(id)\big)^r$.
- **Security**: given g^a, g^r and $H(id)$, the value $e\big(g, H(id)\big)^{ar}$ is indistinguishable from uniform. With $H(id) = g^b$, this is $e(g,g)^{abr}$ given g^a, g^b, g^r: exactly **DBDH**.

Boneh–Franklin: Security

Theorem 4

If DBDH is hard relative to $\mathcal{G}$ and H is modeled as a random oracle, then the construction is a selectively secure IBE.

Boneh–Franklin: Proof by Reduction



Given / To prove

Given: $\Pr\left[\mathrm{IBE}^{\mathsf{sel}}_{\mathcal{A},\Pi}(n) = 1\right] \geq \frac{1}{2} + \epsilon(n)$ for non-negligible ϵ.
To prove: $\Pr\left[\mathrm{DBDH}_{\mathcal{B},\mathcal{G}}(n) = 1\right] \geq \frac{1}{2} + \epsilon(n)/2$.

Boneh–Franklin: Two Cases

Case $\beta = 0$: $T = e(g,g)^{abc}$ (real)

$\mathcal{A}$'s view is exactly as in $\mathsf{IBE}^{\mathsf{sel}}_{\mathcal{A},\Pi}(n)$.

Why

- $mpk = g^a$ and every H-answer (g^b or $g^{x_{id}}$) are uniform in $\mathbb{G}$.
- Every key is correct: $(g^a)^{x_{id}} = H(id)^a$.
- $m_\gamma \cdot T = m_\gamma \cdot e\big(g^a, g^b\big)^c$, so $c^* = E(mpk, id^*, m_\gamma)$ with $r = c$.

So $\Pr[\gamma' = \gamma \mid \beta = 0] \geq \frac{1}{2} + \epsilon(n)$.

Case $\beta = 1$: $T = e(g,g)^z$ (uniform)

c^* carries **no information** about γ.

Why

T is uniform in $\mathbb{G}_T$ and independent of everything else $\mathcal{A}$ sees, so $m_\gamma \cdot T$ is uniform in $\mathbb{G}_T$ whatever γ is: a one-time pad in $\mathbb{G}_T$.

So $\Pr[\gamma' = \gamma \mid \beta = 1] = \frac{1}{2}$.

Boneh–Franklin: Putting It Together

$$\begin{aligned}
\Pr\left[\mathrm{DBDH}_{\mathcal{B},\mathcal{G}}(n) = 1\right] &= \tfrac{1}{2}\Pr[\beta' = 0 \mid \beta = 0] + \tfrac{1}{2}\Pr[\beta' = 1 \mid \beta = 1] \\
&= \tfrac{1}{2}\Pr[\gamma' = \gamma \mid \beta = 0] + \tfrac{1}{2}\Pr[\gamma' \neq \gamma \mid \beta = 1] \\
&= \tfrac{1}{2}\Pr[\gamma' = \gamma \mid \beta = 0] + \tfrac{1}{2}\big(1 - \Pr[\gamma' = \gamma \mid \beta = 1]\big) \\
&\geq \tfrac{1}{2}\big(\tfrac{1}{2} + \epsilon(n)\big) + \tfrac{1}{2} \cdot \tfrac{1}{2} \;=\; \tfrac{1}{2} + \tfrac{\epsilon(n)}{2},
\end{aligned}$$

non-negligible advantage — contradicting the DBDH assumption. □

Remark (running time)

$\mathcal{B}$ runs $\mathcal{A}$ once and does one exponentiation per query: PPT.

Remark (full security)

Without the commitment to id^*, $\mathcal{B}$ **guesses** which of $\mathcal{A}$'s $q(n)$ H-queries is id^* and plants g^b there, as in RSA-FDH: the advantage drops to roughly $\epsilon(n)/q(n)$.

IBE $\Rightarrow$ Digital Signatures

Question: given a secure IBE (S, K, E, D), how would you build a signature scheme?

Construction

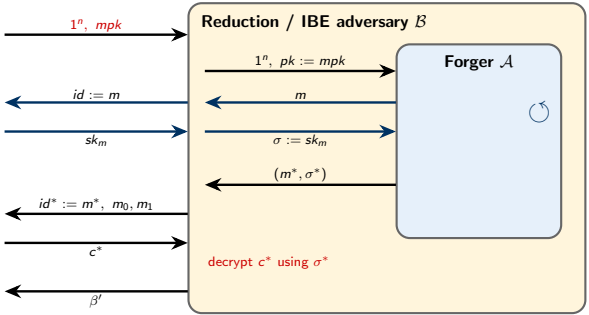
Given a secure IBE (S, K, E, D):

- Gen(1^n): $(mpk, msk) \leftarrow S(1^n)$; $pk := mpk$, $sk := msk$.
- Sign(sk, m): output $\sigma \leftarrow K(msk, m)$.
- Vrfy(pk, m, σ): for a random h from the IBE message space, output 1 iff $D\big(\sigma,\ E(mpk, m, h)\big) = h$.

Intuition

The IBE secret key for identity m *is* the signature on m. Verification is an encrypt-then-decrypt consistency check.

Proof: Signature Forger ⇒ IBE Adversary

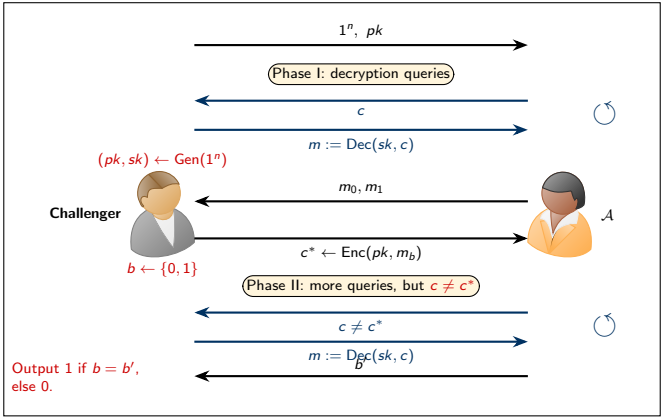


Why it works

$\mathcal{A}$ never asked for m^*, so $\mathcal{B}$ never queried $K(msk, m^*)$: its challenge on $id^* = m^*$ is legal. A valid σ^* decrypts correctly (that is what Vrfy checks), so $\mathcal{B}$ learns m_β and wins.

$\mathrm{PubK}^{\mathrm{cca}}_{\mathcal{A},\Pi}$: Recall

$\mathrm{PubK}^{\mathrm{cca}}_{\mathcal{A},\Pi}(n)$



CCA Security: Recall

Game $\mathsf{PubK}^{\mathsf{cca}}_{\mathcal{A},\Pi}(n)$

1. $(pk, sk) \leftarrow \mathsf{Gen}(1^n)$. $\mathcal{A}$ is given 1^n, pk, and **oracle access** to $\mathsf{Dec}(sk, \cdot)$.
2. $\mathcal{A}$ outputs m_0, m_1 with $|m_0| = |m_1|$.
3. $b \leftarrow \{0,1\}$, $c^* \leftarrow \mathsf{Enc}(pk, m_b)$; c^* is given to $\mathcal{A}$, which may **not** query $\mathsf{Dec}(sk, \cdot)$ on c^*.
4. $\mathcal{A}$ outputs b'. Output 1 if $b = b'$, and 0 otherwise.

Π is **CCA-secure** if for all PPT $\mathcal{A}$: $\Pr\left[\mathsf{PubK}^{\mathsf{cca}}_{\mathcal{A},\Pi}(n) = 1\right] \leq \frac{1}{2} + \mathsf{negl}(n)$.

- **CCA1** (weaker): decryption queries only *before* the challenge. The game above is **CCA2**.
- Achieving CCA security is **much harder in the PKE setting**.

IBE $\Rightarrow$ CCA1-Secure PKE

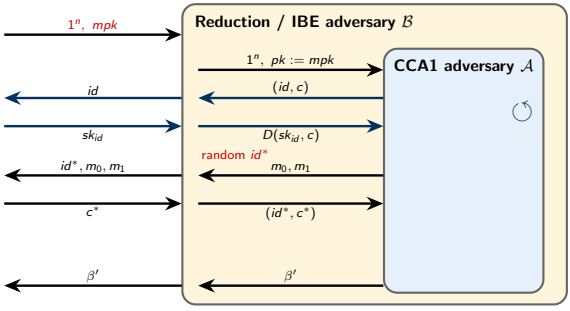
Let's start with CCA1.

Question: given a secure IBE (S, K, E, D), how would you build a CCA1-secure PKE?

Construction

- $\mathrm{Gen}(1^n)$: $(mpk, msk) \leftarrow S(1^n)$; $pk := mpk$, $sk := msk$.
- $\mathrm{Enc}(pk, m)$: sample a **random identity** id; output the ciphertext $\big(id,\ E(mpk, id, m)\big)$.
- $\mathrm{Dec}(sk, (id, c))$: output $D\big(K(msk, id),\ c\big)$.

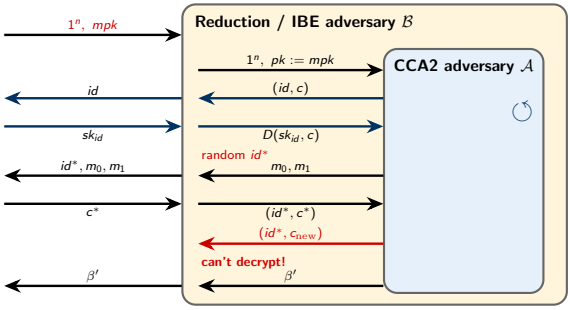
Proof: CCA1 Adversary ⇒ IBE Adversary



Why it works

Each decryption query (id, c) is answered with a key query for id: $\mathcal{B}$ **can decrypt**. The challenge uses a *fresh random* id^*, chosen after all of $\mathcal{A}$'s queries, so $id \neq id^*$ except with negligible probability.

What Is the Problem in Getting CCA2?



The problem

After the challenge, $\mathcal{A}$ can make up new ciphertexts (id^*, c_{new}) for the identity id^*. $\mathcal{B}$ may not ask for sk_{id^*}, so it cannot answer them.

How Can We Fix This?

- How can we **prevent** $\mathcal{A}$ from asking for decryptions of new ciphertexts (id^*, c_{new}) under the challenge identity?

Idea

Replace the identity with the verification key of a signature scheme.

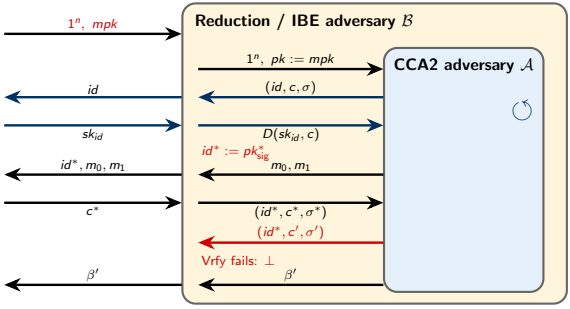
IBE $\Rightarrow$ CCA2-Secure PKE

Use a **strongly unforgeable** signature scheme $(\mathsf{Gen}_{\mathsf{sig}}, \mathsf{Sign}, \mathsf{Vrfy})$.

Construction

- $\mathsf{Gen}(1^n)$: $(mpk, msk) \leftarrow S(1^n)$; $pk := mpk$, $sk := msk$.
- $\mathsf{Enc}(pk, m)$: sample $(pk_{\mathsf{sig}}, sk_{\mathsf{sig}}) \leftarrow \mathsf{Gen}_{\mathsf{sig}}(1^n)$, and
  1. set $id := pk_{\mathsf{sig}}$;
  2. $c \leftarrow E(mpk, id, m)$;
  3. $\sigma \leftarrow \mathsf{Sign}(sk_{\mathsf{sig}}, c)$;
  4. output the ciphertext (id, c, σ).
- $\mathsf{Dec}(sk, (id, c, \sigma))$: if $\mathsf{Vrfy}(id, c, \sigma) = 1$, output $D\big(K(msk, id),\, c\big)$; else output $\perp$.

CCA2-Secure PKE: Why It Works



$\mathcal{B}$ samples $(pk^*_{\text{sig}}, sk^*_{\text{sig}})$ itself and uses $id^* := pk^*_{\text{sig}}$; the challenge is (id^*, c^*, σ^*). A new ciphertext $c_{\text{new}} = (id^*, c', \sigma')$ has $(c', \sigma') \neq (c^*, \sigma^*)$ and needs to be decrypted only if $\text{Vrfy}(id^*, c', \sigma') = 1$: $\mathcal{B}$ safely returns $\perp$ if verification fails. If it succeeds, (c', σ') is a **forgery** for the signature scheme. **One-time** strong unforgeability suffices.

Summary

- **IBE**: the recipient's identity is the public key. Security ($\mathrm{IBE}_{\mathcal{A},\Pi}(n)$): keys for any identities except the challenge id^*; **selective** security fixes id^* up front.
- **Pairings**: bilinear maps (DDH easy, CDH hard) and the **DBDH** assumption give one-round three-party key exchange (Joux) and an IBE construction.
- **IBE** $\Rightarrow$ **signatures**: the secret key for identity m is the signature on m; a forgery decrypts the IBE challenge.
- **IBE** $\Rightarrow$ **CCA**: fresh random identities give CCA1; a signature verification key as the identity (and a signature on the ciphertext) gives **CCA2**.